2.1 → Ratemaking principles, considerations, + adjustments

2.1.0 → overview + 2.1.1 → Ratemaking principles + considerations

→ Ratemaking is the process of determining the premium so that the fundamental insurance equation is balanced

Premium = Losses + LAE + UW Expenses + UW Profit

→ Ratemaking principles → Rates charged by insurance companies should be developed by following these four principles:

- → Principle 1 → A rate is an estimate of the expected value of future costs
- → Principle 2 → A rate provides for all costs associated w/ the transfer of risk (ie balance at aggregate level)
- → Principle 3 → A rate provides for the costs associated w/ an individual risk transfer (ie balance at individual level)
- → Principle 4 → A rate is reasonable + not excessive, inadequate, or unfairly discriminatory if it is an actuarially sound estimate of the expected value of all future costs associated w/ an individual risk transfer

→ Ratemaking considerations → Things to take into account when developing ratemaking methodologies

→ Exposure unit, data, organization of data, influences on data, adjustments to data, Homogeneity, credibility, classification plans, individual risk rating, actuarial judgement

Statement of principles regarding P&C ratemaking
⇒ A rate change based solely on profit violates the first three principles

→ Influences on data → more things to be considered

→ policy provisions, mix of business, operational changes, other influences

→ Evaluating + adjusting ratemaking data → Adjustments to make + ensure the data reflects anticipated future costs accurately

→ shock losses, reinsurance, benefit + rate adjustments, development, trends, expenses, profit + contingency provision

2.1.2 → Shock losses

→ Overview → historical losses need several adjustments before they can be used for ratemaking, including:

- → Replacing extraordinary events in the historical data w/ long term expectations
- → Adjusting losses for changes in coverage or benefits, including changes required by law
- → Developing immature losses to their ultimate settlement value
- → Trending losses to reflect the benefit + cost levels expected during the period the rates will be in effect

→ There isn't a specific order in which these adjustments need to be made, but the order should still be considered so that the proper adjustments are made. For instance, if the long term expectation for extraordinary losses has already been trended to the period the rates will be in effect, this expectation should not be added to historical losses until these losses have also been trended to the same period.

→ Pricing based on occurrences of shock losses

- → If no shock losses occur, prices will an underestimation of losses compared to the true price when shock losses are considered ⇒ ↓ indicated rate
- → If shock losses do occur, prices will be an overestimation of losses — — — — — — — — ⇒ ↑ indicated rate

→ ☆ Same for CATS

→ Adjusting for shock losses (steps)

→ 1) options include → Excluding shock losses entirely
- → Capping all losses based on the minimum amount of insurance offered, typically referred to as the basic limit, or the limit associated w/ the base rate
- → Capping losses at a different large loss threshold

→ If shock losses are removed entirely, the remaining losses are referred to as non-shock losses.

→ If shock losses are capped at a threshold, then only the portion of losses above that threshold, referred to as excess losses, will be removed. The remaining losses are known as non-excess losses.

→ 2) Include provision
→ First need to consider two conflicting goals though when determining the threshold
- → 1) Include as many losses as possible
- → 2) minimize volatility in the ratemaking analysis

→ Determining shock loss provision

→ once a threshold is set for excluding or capping shock losses, a provision for shock losses can be calculated using the following steps:

1) Separate all losses into excess losses + non-excess losses
2) Sum the excess losses
3) Sum the non-excess losses
4) Calculate the ratio of excess losses to non-excess losses
5) Calculate the excess loss loading factor as 1 plus the ratio found above.

→ In general: $\text{ratio} = \frac{\sum \text{losses adjusting for}}{\sum \text{non (...) losses}}$

$\text{factor} = 1 + \text{ratio}$

$\text{year}_i \text{ adj losses} = \text{non (...) losses} \times \text{factor}$

→ Then this factor is multiplied to the historical non-excess losses to account for the long term average expected shock loss.

2.1.3 → Catastrophe data

→ Trend before calculating BEFORE calculating shock loss load factor
↳ to avg accident date in future

→ Adjusting for catastrophes

→ To adjust for catastrophes when projecting future losses for ratemaking, actual catastrophe losses can be replaced by the average expected cat loss.

→ The method for determining the expected cat loss can vary by the type of insurance + the type of cat loss. Cat losses are typically broken down into the following two types:

1) Non-modeled cat losses
2) modeled cat losses

→ non-modeled cat analysis is used for events that occur somewhat regularly over a period of many year, like hail storms.

→ In contrast, modeled catastrophe analysis is used for events that occur very infrequently but that result in high severity claims, like hurricanes or earthquakes

→ not covered, but if given, need to use ⇒ Total cat provision = non-modeled cat pure premium + modeled cat loss pure premium

→ Determining non-modeled cat loss provision

→ Method 1: cat loss loading factor

→ one method for setting the non-modeled cat loss provision is to follow the same process used for the shock loss provision.

1) Sum the cat losses
2) Sum the non-cat losses
3) Calculate the ratio of cat losses to non-cat losses
4) Calculate the cat loss loading factor as 1 plus the ratio found above

→ Then, multiply non-cat losses used in the ratemaking analysis by the factor above to incorporate the long-term average expected non-modeled cat loss.

→ Method 2: Cat loss pure premium

→ A different method for incorporating non-modeled cat losses is to develop a pure premium or loss ratio just for the non-modeled cat exposure.

→ For instance, assume we need the non-modeled cat loss pure premium for a homeowners' book of business. Using the amount of insurance years (AIY), which is the total amount of insurance for all policies inforce during the calendar year, the process for determining the pure premium is:

1) Calculate the ratio of non-modeled cat losses to AIY for each year in the experience
2) Average the cat-to-AIY ratio for all years
3) Apply the non-modeled cat provision per AIY to the average AIY per exposure expected in the future to get the non-modeled cat pure premium

→ in general: $\frac{\text{cat loss}}{\text{AIY}} \times \frac{\text{AIY}}{\text{Exposures}} = \frac{\text{Cat loss}}{\text{Exposures}} = \text{Cat PP}$

$\approx \frac{\text{claim count}}{\text{Exposure}} \times \frac{\text{losses}}{\text{claim count}} = \frac{\text{losses}}{\text{Exposure}} = \text{PP}$

(similar to → (freq × sev))

→ Appropriate to use a large loss loading factor when there has been a large loss, duh...

⇒ can look at the excess loss for each AY relative to the historical avg excess loss (if given)

⇒ goal of load is to smooth/average out effect of these losses over years b/c they are sporadic + volatile

→ Trend losses BEFORE calculating shock loss factor

→ XS loss is only is shock losses, particularly the losses above the XS loss threshold

⇒ Non XS losses includes the shock losses under the XS loss threshold

→ Methods to select XS loss threshold ⇒ Actuarial judgement + select a quantile from the loss distribution

→ XS loading factor → rather than calculating Total XS losses (a) + total non-XS losses (b), + doing $\frac{a}{b}$

⇒ just calculate XS loss (a) + TOTAL loss (c), + do $\frac{a}{c-a}$

2.1.4 → Reinsurance

→ Types → Proportional (ie quota-share, e.g. 70% / 30%) (70% → us, 30% → them)

Include in ratemaking analysis? → no / yes (ie reduce losses + premium proportionally OR include net cost of reinsurance as an expense item)

→ non-proportional → excess-of-loss

⇒ b/c AFFECTS final loss ratio

2.2 → Rate Benefit + Changes

2.2.1 → Examining effects of rate + benefit changes

→ Effects of changes

→ Direct effects are direct + obvious impacts on premium or losses resulting from changes in rates or benefits. Examples include:
- → ↑ rates ⇒ ↑ premium
- → ↓ policy limit ⇒ ↓ losses
- → excluding certain coverages ⇒ ↓ losses

→ Indirect effects are less straightforward

→ Measuring effects → Examining impact of various changes on premium is just change by policy, easy!
→ for losses, restating each claim individually is the ideal method

→ Current-Rate level simply demonstrated

→ Recall that when using past data for ratemaking, it is important to adjust the data to what is expected in the future when rates will be in effect.

→ For instance, assume that all insurance policies during the historical period were written at a rate of $100. After the historical period, a rate increase of 10% went into effect, which means the current rate in effect is $110.

→ now, assume the indicated rate for the future ratemaking period is $115. If the 10% rate increase that went into effect after the historical period is not considered, then the historical rate of $100 will be compared to the indicated rate of $115. This will imply the current rates should increase by 15%. However, this would cause rates to actually increase to 110(1.15) = 126.5, which is higher than what was indicated.

→ By first restating the historical premium to the current rate level of $110, the actual rate increase needed for an indicated rate of $115 can be calculated: $\frac{115}{110} - 1 = 0.0455$

2.2.2 → Adjusting historical premium: Extension of exposures method

→ Extension of exposures

→ The extension of exposures method entails rerating each policy to restate the historical premium as if it had been charged under the current rates. This process can be divided into two steps:

1) Rerate every policy using the current rates. This means we take the rating characteristics of each policy from the historical period + recalculate the premium based on the current rates
2) Recalculate the EP for each period using the updated rates

→ CRL is simply using the current rates on all policies

⇒ $\text{on-level factor} = \frac{\text{EP @ CRL}}{\text{CY EP}}$ (↳ raw, original EP)

→ challenges → necessary detailed data may not always be available
- → Involves extensive calculations
- → Assumptions are made for new rating variables when no data is available

2.2.3 → Adjusting historical premium: Parallelogram method + 2.2.4 → Adjusting historical losses

→ overview → Easier, but less accurate than extension of exposures method

☆ → Assumes: 1) policies are written uniformly over time + 2) typically applied at an aggregate level using a series of overall avg rate changes

⇒ seasonality violates assumption ↳ to get around this, can use smaller time intervals, such as quarters

→ Parallelogram method standard calculations

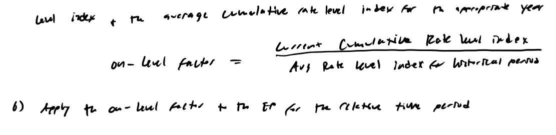
→ To summarize, the parallelogram method requires the following steps:

☆

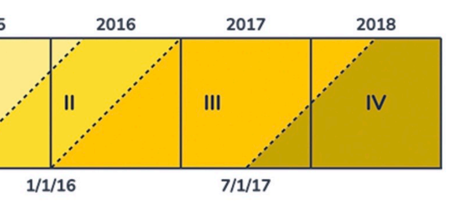
1) Determine the timing + amount of the rate changes
2) For each rate level group, calculate the cumulative rate level index
3) For each time period, calculate the portion of premium that corresponds to each rate level group
4) For each time period, calculate the weighted average cumulative rate level index
5) For each time period, calculate the on-level factor as the ratio of the current cumulative rate level index + the average cumulative rate level index for the appropriate year

$$\text{on-level factor} = \frac{\text{Current Cumulative Rate level index}}{\text{Avg Rate level index for historical period}}$$

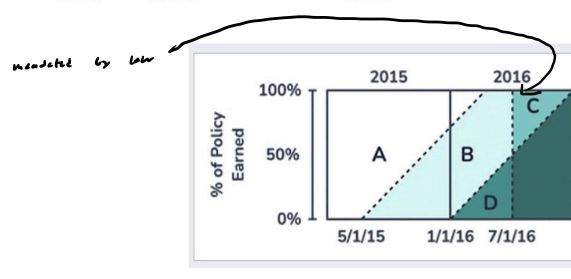
6) Apply the on-level factor to the EP for the relative time period



→ Rate changes mandated by law



→ Policy Year



(same steps for PY)

...

☆ (same for losses too)

→ 1) Draw picture

→ 2)

(1) Group	(2) Eff date	(3) Rate change	(4) = 1 + (3) Rate index	(5) Cumulative RLI
1		--	1.00	↑↑↑
2		5%	1.05	
3		-2%	0.98	Product

(6) Portion: CY1, CY2, CY3 (Σ = 1, Σ = 1, Σ = 1)

(7) weighted average RLI: CY1 ○, CY2 ○, CY3 ○ — sumproduct[(5), (6)]

(8) on-level factor: CY1 ○, CY2 ○, CY3 ○ — $\frac{(5)\text{ last}}{(7)}$

→ On-leveling WP

→ So far, we've covered how to on-level EP. When on-leveling WP for policy years, the approach will be the same as on-leveling EP since EP for a PY is equal to WP. However when on-leveling WP for calendar years, the approach will be slightly different.

⇒ company rate changes = ⁞ vertical + rate changes mandated by law = ⁄ diagonal

→ Diagonals + verticals only change for CY WP

⇒ Law changes (diagonals) go backwards in time (think about what policies it affects)

⇒ law changes to premium apply to the CY when the law was enacted (might have to take this into account)

→ Parallelogram method → larger cumulative average rate level ⇒ smaller on-level factor ⇒ less EP

→ weighted average RLE / not cumulative RLI

2.3 → Development

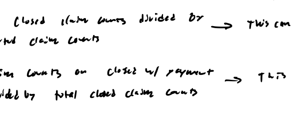
2.3.1 → Development triangles

→ list of the items a reserving actuary may discuss w/ the various departments of an insurer.

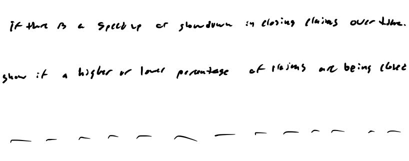
→ Claims → Addressing unpaid case guidelines, changes in reserving procedures, caseload changes, the process of establishing a claim file, reopened claims, shifts in settlements or adjustments of claims, + fraud safeguards.

→ Underwriting → Inquiries about changes in business composition, underwriting guidelines, large risks, self-insured coverages, program tracking, + pricing of new programs

→ Example →



Accident Year	Cumulative Paid Claims by Age ($US Millions)			
	12 Months	24 Months	36 Months	48 Months
2019	102	135	161	160
2020	100	121	152	
2021	108	150		
2022	110			



Accident Year	Valuation Dates by Age			
	12 Months	24 Months	36 Months	48 Months
2019	12/31/19	12/31/20	12/31/21	12/31/22
2020	12/31/20	12/31/21	12/31/22	
2021	12/31/21	12/31/22		
2022	12/31/22			

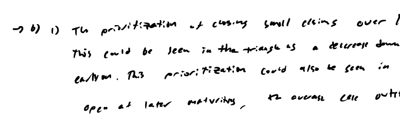
☆ pay attention to what's given

→ There are several different development triangles of ratios + average values that can be used for diagnostics. Some common examples + what they can be used for include (these are just examples, don't need to memorize)

- → Paid losses divided by reported losses → This can show if there is a speedup or slowdown in closing claims over time. It can also show changes in case reserve adequacy
- → Total closed claim counts divided by reported claim counts → This can show if there is a speedup or slowdown in closing claims over time.
- → Claim counts on closed w/ payment divided by total closed claim counts → This can show if a higher or lower percentage of claims are being closed w/ pay.
- → Claim counts on closed w/o payment divided by total closed claim counts → — — — — — — — — — — w/o pay.
- → Average case reserves (case reserves divided by outstanding claim counts) → This can show severity trends, if there is a speedup or slowdown in closing small claims relative to large claims, or changes in case reserve adequacy
- → Average paid on closed claims (cumulative paid losses on closed claims divided by claim counts closed w/ payment) → This can show if there is a speedup or slowdown in closing small claims relative to larger claims. It can also show severity trends
- → Average reported (reported losses divided by reported claim counts) → This can show changes in case reserve adequacy.

→ Looking for stability down columns (ie across AYs for the same maturity period)

→ Examples →



(a) Identify three changes in an insurance company's internal environment that could distort the paid or reported development patterns.

(b) Briefly describe how each change identified in part (a) above may be observed in a diagnostic triangle.

→ a) 1) Prioritization of closing small claims over large claims
2) Paying out claims faster
3) Change in policy limits

→ b) 1) The prioritization of closing small claims over large claims would show up in the average paid on closed claims triangle. This could be seen in the triangle as a decrease down the columns for early maturities since smaller claims are being closed earlier on. This prioritization could also be seen in the average case outstanding triangle. Since larger claims would be left open at later maturities, the average case outstanding would increase down the columns.

2) If claims are paid faster, the paid to reported claims triangle could be looked at. This triangle should show an increase down the columns since paid claims would increase as claims are settled more quickly, while reported claims would not change.

3) A change in policy limits could affect the average paid triangle. If policy limits are increased, this could be seen as an increase down the columns, as the maximum amount paid on a claim would be higher, which could increase the average paid amounts.

→ Note → There are many possible changes that could be identified in part (a), such as:
- → An increase or decrease in case reserve adequacy
- → A change in the mix of business
- → A change in underwriting rules
- → A reinsurance program change
- → Starting a fraud detection department
- → A change in how claims are processed, such as introducing a new technology
- → Increased risk control (safety programs)

→ For the exam, it is helpful to select changes that impact development patterns more directly, as this makes it easier to answer part (b)